AI ACADEMY · LEVEL 2 · GRADE 7 · WEEK 11 · WORKBOOK

Training and Testing

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Transparent Book Recommender

THE CONNECTED WEEK

One connected split-audit record.

Explain roles, audit the A–F plan, repair related-frame overlap and reconsider separation for new routes.

Keep one habit: Retain exact assignments, row/group counts, pairwise overlaps, versions and limitations.

MAIN LESSON · SECTION 1

Recall: reconnect earlier ideas

> Retrieve before opening notes.

Retrieve and explain

- Try the recall before opening notes. Then use a different colour or a labelled second line to correct your explanation.

- Without opening the old lesson, explain one key idea from Human Classifier Simulation. Give an example and a mistake that the idea helps you avoid.

- Without opening the old lesson, explain one key idea from Operational Labels. Give an example and a mistake that the idea helps you avoid.

- Without opening the old lesson, explain one key idea from Loops and Stopping. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

My remembered idea, example and correction

My remembered idea, example and correction

▼ **Earlier wording, alternatives and picture transcripts**

Human Classifier Simulation

Operational Labels

Loops and Stopping

W1 · Explain training, validation and test roles

> Distinguish final evaluation from repeated development choices.

Your evidence record

- A–F reference: training A1 A2 A3 B1 B2 C1; validation D1 D2; test E1 E2 F1 F2. Group equals the trip letter, not the full frame ID. Source inventory has twelve distinct frames from six trips. The claim concerns new trips on a familiar route. The group script compares the group sets for overlap and counts distinct groups; it does not audit row completeness or fit a model.

- Explain what **training**, validation and **test set**s are used for. What changes if you repeatedly choose a model using its test score?

> **My reasoning and evidence**

MAIN LESSON · SECTION 13

W2 · Audit group and frame counts

> Use the supplied assignment and complete group-list script.

Your evidence record

- A–F reference: training A1 A2 A3 B1 B2 C1; validation D1 D2; test E1 E2 F1 F2. Group equals the trip letter, not the full frame ID. Source inventory has twelve distinct frames from six trips. The claim concerns new trips on a familiar route. The group script compares the group sets for overlap and counts distinct groups; it does not audit row completeness or fit a model.

- For the worked A through F plan, give **training**, validation and test counts first in **group**s and then in rows. Interpret [] from the code and state one problem it cannot detect.

```
train_groups = {"A", "B", "C"}
valid_groups = {"D"}
test_groups = {"E", "F"}
shared = ((train_groups & valid_groups)
          | (train_groups & test_groups)
          | (valid_groups & test_groups))
print(sorted(shared))
print(len(train_groups), len(valid_groups),
      len(test_groups))
```

> **My reasoning and evidence**

MAIN LESSON · SECTION 14

W3 · Repair trip A crossing roles

> Unique IDs and a seed cannot remove a shared source.

Your evidence record

- A–F reference: training A1 A2 A3 B1 B2 C1; validation D1 D2; test E1 E2 F1 F2. Group equals the trip letter, not the full frame ID. Source inventory has twelve distinct frames from six trips. The claim concerns new trips on a familiar route. The group script compares the group sets for overlap and counts distinct groups; it does not audit row completeness or fit a model.

- A1 and A2 are in **training** and A3 is in test. Explain why unique frame IDs and a fixed random seed do not fix this. Propose a repair and what to record.

> **My reasoning and evidence**

MAIN LESSON · SECTION 16

W5 · Change the split when the claim becomes new routes

> State the two-route limitation rather than claiming a complete three-way evaluation.

Your evidence record

- A model will be evaluated on new routes. Route North has trips A and B; route South has C and D. A proposed split trains on A and C and tests on B and D. Explain the problem and give a route-based alternative.

  Is this enough for a strong performance claim?

My reasoning and evidence

MAIN LESSON · SECTION 4

W4 · Audit and repair independently

> Keep one first attempt before hints or feedback.

Read and explain independently

Eight frames come from trips P (P1 P2 P3), Q (Q1 Q2), R (R1), S (S1 S2). A proposed split is **training** P1 Q1 R1; validation P2 S1; test Q2 P3 S2. Identify shared **group**s for each pair of sets.

Repair it by assigning P and Q to **training**, R to validation and S to test. List all frames in each role and give row and **group** counts. State one limit of the repaired plan.

My independent explanation, trace and limits

Keep a fair record

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

▼ Earlier wording, alternatives and picture transcripts

Independent case: Audit and repair independently

MAIN LESSON · SECTION 9

Project · Keep the decision and its evidence

> Add this week's change to your existing project record.

Your evidence record

Keep final user profiles hidden until weights and thresholds are fixed.

The project change and **evidence** I will keep

> **Write or draw your reasoning and evidence.**

MAIN LESSON · SECTION 17

Feedback · Repair and return

> Keep the first attempt, the repair and a changed-case return.

Your evidence record

Evidence	My record
Worked alone / hint / model	
First step I repaired and why	
New case checked later; date and result	
What I can now explain without notes	
One remaining question	

Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

MAIN LESSON · SECTION 10

Finish · Judge the evidence in your work

> Point to work that supports your confidence level.

Your evidence record

- I can explain the mechanism.
- I can show evidence.
- I can handle a changed case.
- I can name a limit.

Mark each as not yet, with help or independently, and attach supporting work.

Teacher review _____ Next step _____

Reference bank and extra practice

Choose the reference that helps your next step.

▼ **Reference · Supported W2 record**

REFERENCE · SECTION 2

Reference · Supported W2 record

> This is the same W2 attempt. Keep one response record.

Use this when helpful

Use W2: Distinguish frames from **group**s for Splits and related examples. Record the answer once; the lesson and workbook refer to the same attempt.

Given inputs	Procedure I chose	Middle steps	Result and check

> **The step I completed without help, and the hint I needed**

▼ Extra practice · Repair the explanation

REFERENCE · SECTION 3

Extra practice · Repair the explanation

> Explain the deciding fact before discussing the answer.

Use this when helpful

Use the worked case for Splits and related examples. Proposed explanation: “Different row IDs mean independent examples”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

My explanation before partner discussion

▼ Reference · Original learning target

REFERENCE · SECTION 5

Reference · Original learning target

> Use this target to inspect an explanation.

Use this when helpful

A team tests five rule versions on the same ten books and chooses the highest score. It calls the final score an unbiased **unseen** test result. Explain the issue and revise the evaluation plan.

▼ **Extra practice · A complete case**

REFERENCE · SECTION 6

Extra practice · A complete case

> Try the question before checking the worked explanation.

Use this when helpful

A rule is developed and evaluated using the same books A–H.

Why does this not establish **unseen** performance?

Show your trace, calculation or **evidence**

> **Write or draw your reasoning and evidence.**

▼ **Extra practice · A changed case**

REFERENCE · SECTION 7

Extra practice · A changed case

> Keep this practice separate from the reserved independent attempt.

Use this when helpful

You tune a rule using cards 1 to 8 and test it on cards 9 and 10. After seeing card 9's error you change the rule. Is card 9 still untouched test **evidence**?

My practice answer and reason

> **Write or draw your reasoning and evidence.**

Week 11 Test and reflect

▼ **Extra practice · Design a revealing test**

REFERENCE · SECTION 8

Extra practice · Design a revealing test

> Predict before tracing or testing.

Use this when helpful

Create a changed **input**, boundary case or missing-information case. Write the expected answer or safe response before testing.

Test record	My evidence
Changed input or condition	
Expected answer or fallback	
Observed result	
Why the result occurred	

▼ **Reference · Original target response space**

REFERENCE · SECTION 11

Reference · Original target response space

> This teaching case is practice rather than a fresh assessment.

Use this when helpful

A team tests five rule versions on the same ten books and chooses the highest score. It calls the final score an unbiased **unseen** test result. Explain the issue and revise the evaluation plan.

My explanation after practice

Record support used: none, hint or teacher explanation.

▼ Reference · Independent-case response copy

REFERENCE · SECTION 15

Reference · Independent-case response copy

> This duplicates W4. Reuse the same first-attempt record.

Use this when helpful

Eight frames come from trips P (P1 P2 P3), Q (Q1 Q2), R (R1), S (S1 S2). A proposed split is **training** P1 Q1 R1; validation P2 S1; test Q2 P3 S2. Identify shared **group**s for each pair of sets.

Repair it by assigning P and Q to **training**, R to validation and S to test. List all frames in each role and give row and **group** counts. State one limit of the repaired plan.

My reasoning and evidence

▼ Extra practice · Use the worked picture

REFERENCE · SECTION 18

Extra practice · Use the worked picture

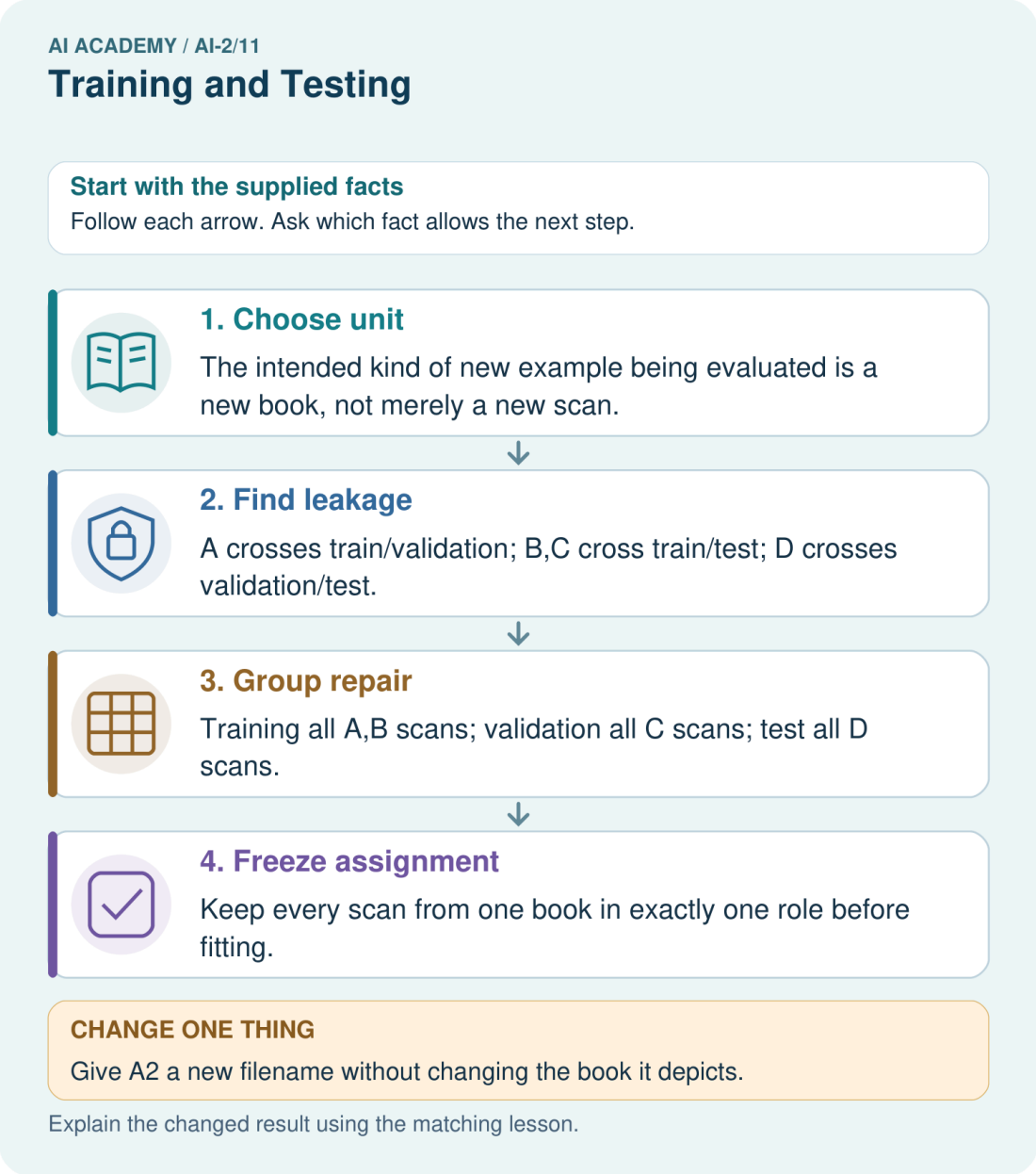
> Keep the explanation and the support you used.

Use this when helpful

A fictional catalogue model learns from scans of book covers and will face previously **unseen** books.

Book A has scans A1,A2; B has B1,B2; C has C1,C2; D has D1,D2.

Proposed **training** A1,B1,C1; validation A2,D1; test B2,C2,D2.



Follow the labelled picture with the matching explanation. Open the vector diagram

How many distinct books are in each repaired role, and why report that as well as scan counts?

> **My picture-based explanation and changed-case prediction**
>
> | |
> |---|
> | |
> | |
> | |
> | |

Support used: alone / hint / worked example. This record is practice **evidence**.

Project connection: **Group** the recommender's related reviews, editions or repeated-user examples according to its intended evaluation claim.

▼ **Reference · Project portfolio record**

REFERENCE · SECTION 19

Reference · Project portfolio record

> Extend your main project response when helpful.

Use this when helpful

Project: Transparent Book Recommender

Current milestone: Make a transparent first version. Keep a trace of how one **input** becomes an **output**; explain the rule or learned component.

Group the recommender's related reviews, editions or repeated-user examples according to its intended evaluation claim.

Evidence to keep	My record
This week's input and deciding step	

Evidence to keep	My record
Prediction and actual result (or labelled paper trace)	
One change since the previous checkpoint and why	
One error or limit and the next test	

> **Explain what this week adds to your project. Reuse last week's record rather than starting again.**

▼ **Extra practice: pictures, repair and feedback**

REFERENCE · SECTION 20

Extra practice: pictures, repair and feedback

> Attempt each question before opening its explanation. Use one browser or paper copy.

First supported question

- A fictional catalogue model learns from scans of book covers and will face previously **unseen** books.
- Which labelled step matches this detail? Keep every scan from one book in exactly one role before **fitting**.
- 1. **Group** repair

- 2. Choose unit
- 3. Find **leakage**
- 4. Freeze assignment
- Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Practice 1: explain the deciding evidence

> **My selected conclusion, deciding fact and explanation**

▼ **Check the first explanation**

After your attempt, compare with the supported explanation: Freeze assignment Keep every scan from one book in exactly one role before **fitting**. Different scan IDs do not make examples independent books.

A model may recognise repeated cover details from another view. **Group**ing by book matches the intended claim about new books and makes the split inspectable.

The repaired split has four **training**, two validation and two test scans, but only two, one and one books respectively.

Changed-case question

- Changed-case practice: Give A2 a new filename without changing the book it depicts. Which conclusion is supported by the supplied facts?
- 1. Its **group** is still A; the original related examples still cross the split.
- 2. This one example guarantees correct results for every future **input**.

- 3. We can decide the result without examining the changed **input** or the procedure.

- Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Practice 2: explain the deciding evidence

> **My selected conclusion, deciding fact and explanation**
>
> | |
> |---|
> | |
> | |
> | |
> | |

▼ **Check the changed case**

After your attempt, compare with the supported explanation: Its **group** is still A; the original related examples still cross the split. Renaming a row does not remove the relationship between scans.

Repair and save your work

Review Human Classifier Simulation (ai-2/10). Goal: Construct a classifier with a stated **input** format, selected rule and recorded **output**.

Short practice: Why are J and M initially treated differently even though both are illustrated mysteries?. Return to this week and explain what you changed.

Review Operational Labels (ai-2/8). Goal: Apply a category definition using explicit **evidence** and boundary rules. Short practice: Why can we not resolve the disagreement merely by taking a majority of reviewers?.

Return to this week and explain what you changed.

Review Loops and Stopping (ai-2/5). Goal: Trace variable values before and after each loop iteration. Short practice: Why is report 0 correct for the original loop, and why is equality-only stopping risky here?.

Return to this week and explain what you changed.

The browser lab lets you edit the supported Python examples already included in this lesson. Predict first, run, inspect the actual **output** or error, change one **input**, then run again.

Keep code, **prediction**s, actual results and a reasoned revision together. On paper, label your result as a trace rather than an execution.

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

Drafts stay in this browser when storage is available; clear drafts on a shared device.

> **What I repaired, the evidence I kept and the next test**
>
> | |
> |---|
> | |
> | |
> | |
> | |

▼ **Earlier wording, alternatives and picture transcripts**

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Repair a missing building block

Predict, run and debug

Keep your completed work

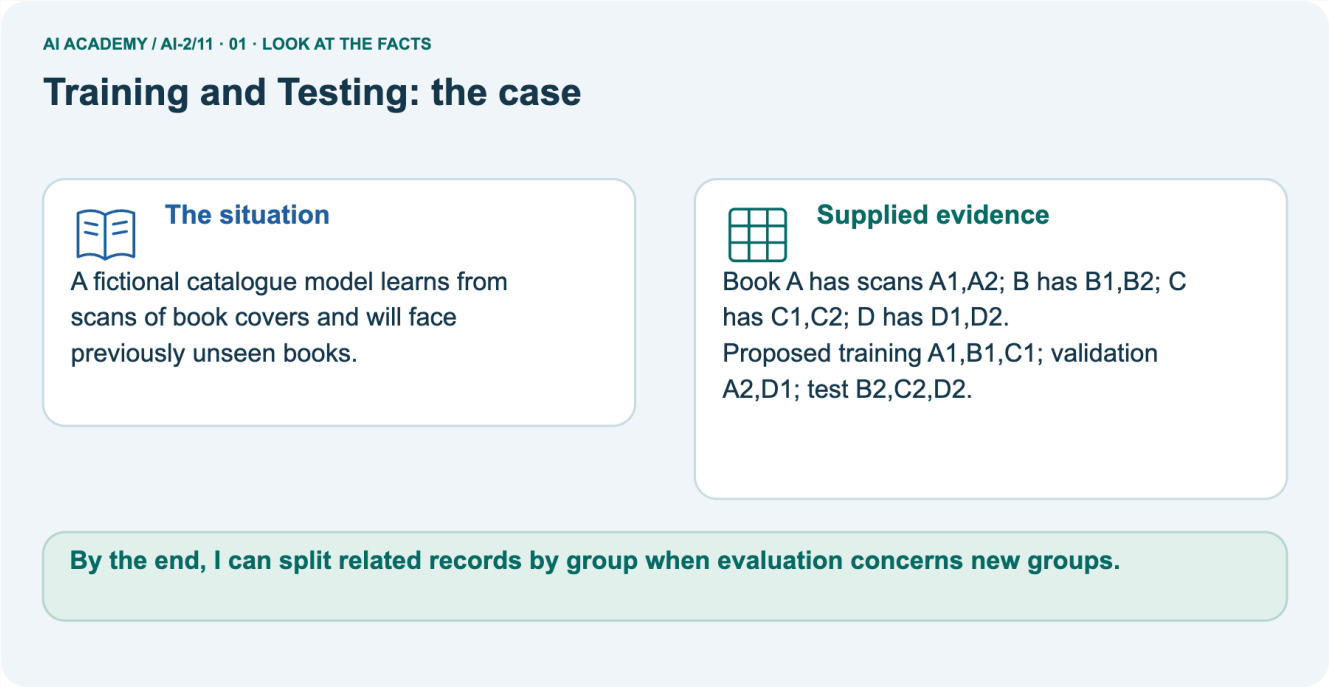
▼ **Reference · Six-picture response sheets**

REFERENCE · SECTION 21

Reference · Six-picture response sheets

> Choose the image that helps rather than repeating all six.

01. Training and Testing: the case



Follow the labelled picture with the matching explanation. Open the vector diagram

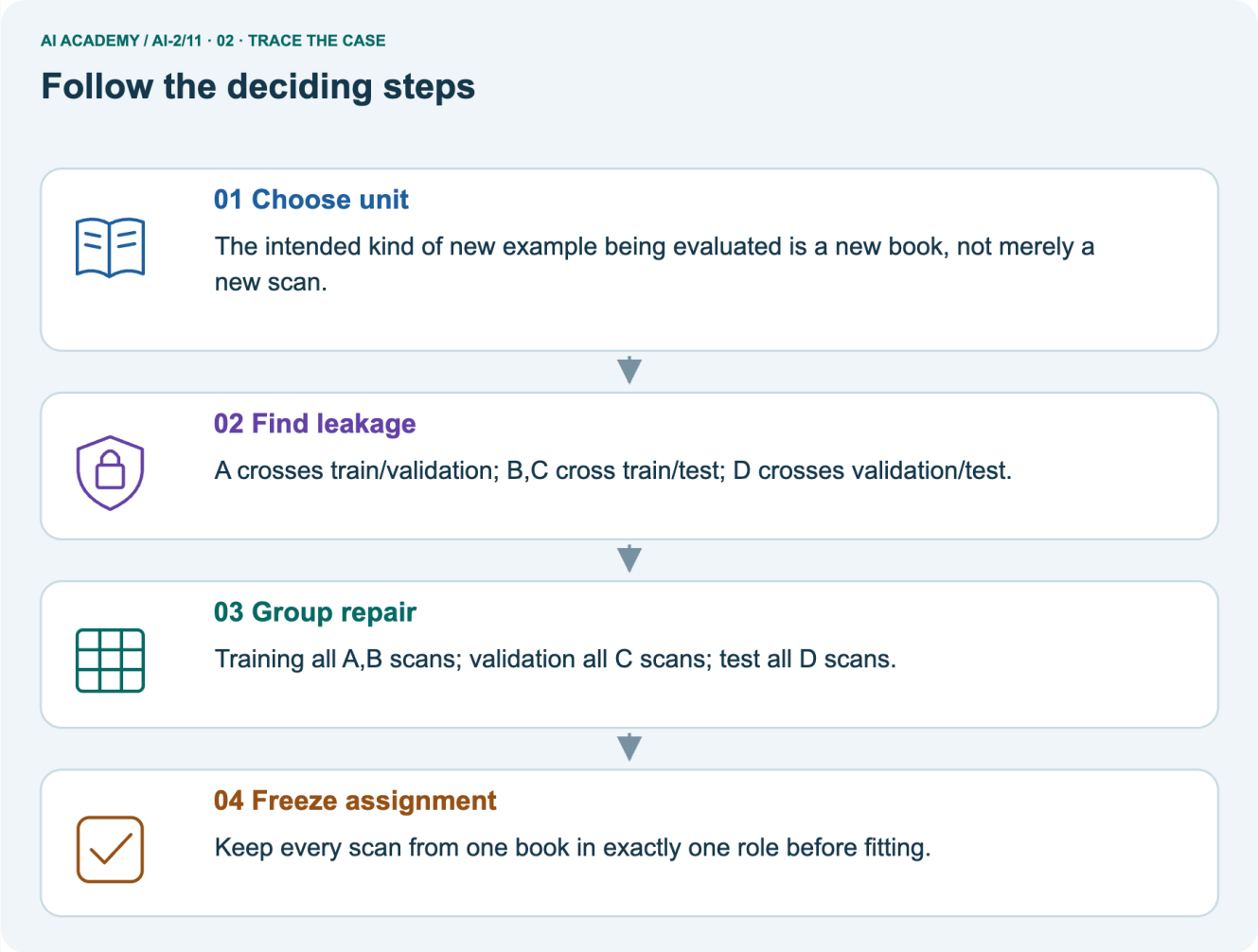
A fictional catalogue model learns from scans of book covers and will face previously **unseen** books. Supplied facts: Book A has scans A1,A2; B has B1,B2; C has C1,C2; D has D1,D2.

Proposed **training** A1,B1,C1; validation A2,D1; test B2,C2,D2.

Point to the given fact that matters most. Explain why it matters.

My picture explanation, deciding evidence and one limit

02. Follow the deciding steps



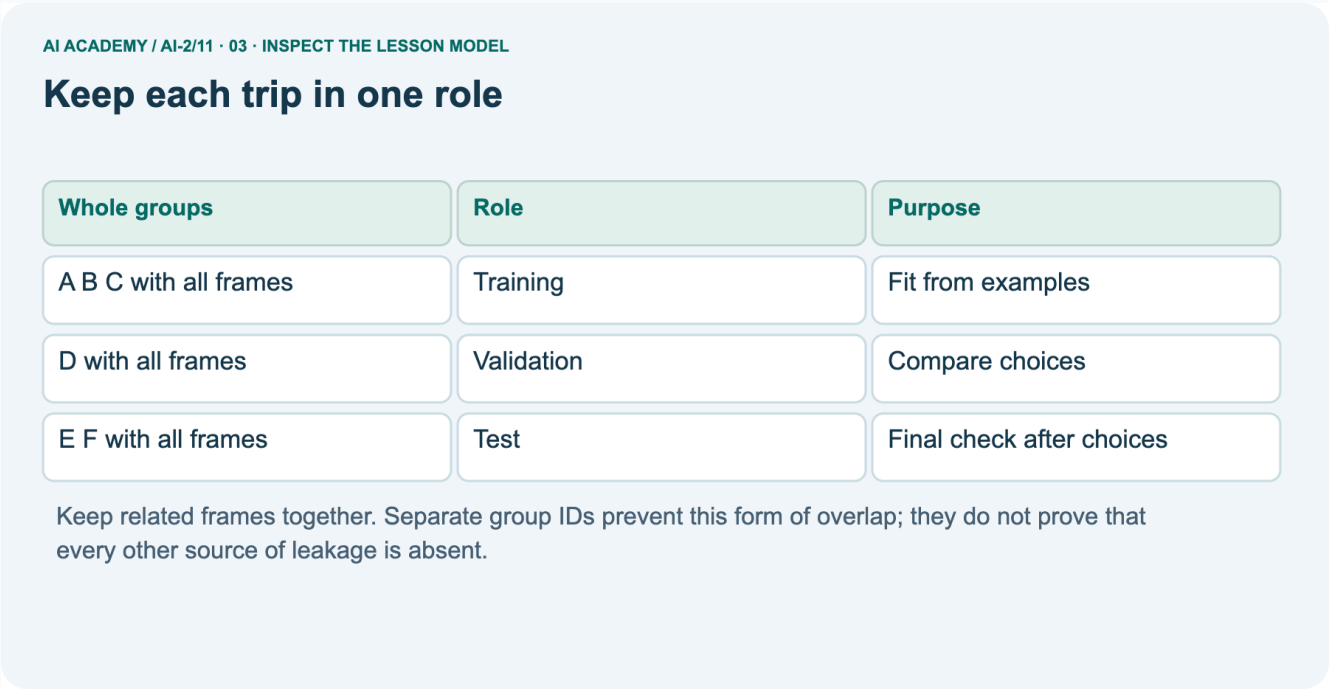
Follow the labelled picture with the matching explanation. Open the vector diagram

Choose unit: The intended kind of new example being evaluated is a new book, not merely a new scan. → Find **leakage**: A crosses train/validation; B,C cross train/test; D crosses validation/test. → **Group** repair: **Training** all A,B scans; validation all C scans; test all D scans. → Freeze assignment: Keep every scan from one book in exactly one role before **fitting**.

Point to each numbered stage and explain which recorded **input** it uses.

My picture explanation, deciding evidence and one limit

03. Keep each trip in one role



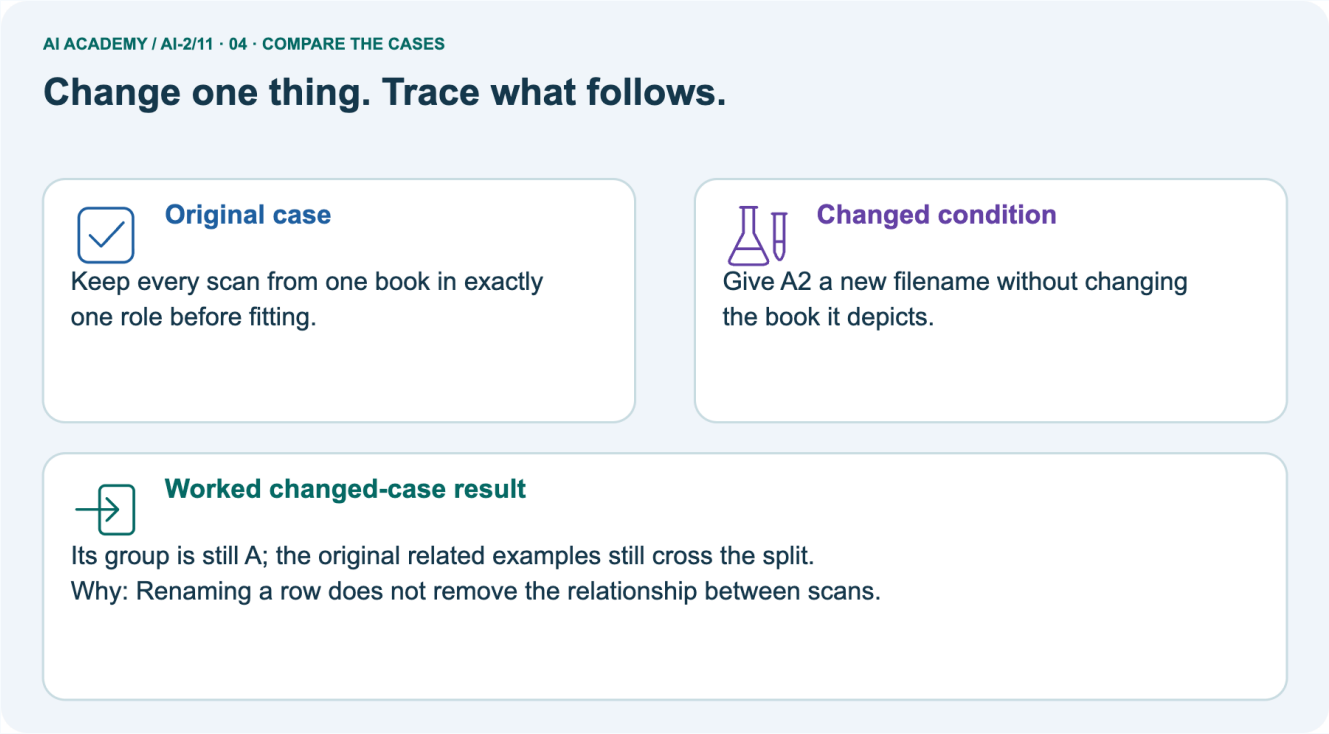
Follow the labelled picture with the matching explanation. Open the vector diagram

Keep each trip in one role. A B C with all frames | **Training** | Fit from examples. D with all frames | Validation | Compare choices. E F with all frames | Test | Final check after choices

Read the labels and values. Explain the relationship the model shows.

My picture explanation, deciding evidence and one limit

04. Change one thing. Trace what follows.



Follow the labelled picture with the matching explanation. Open the vector diagram

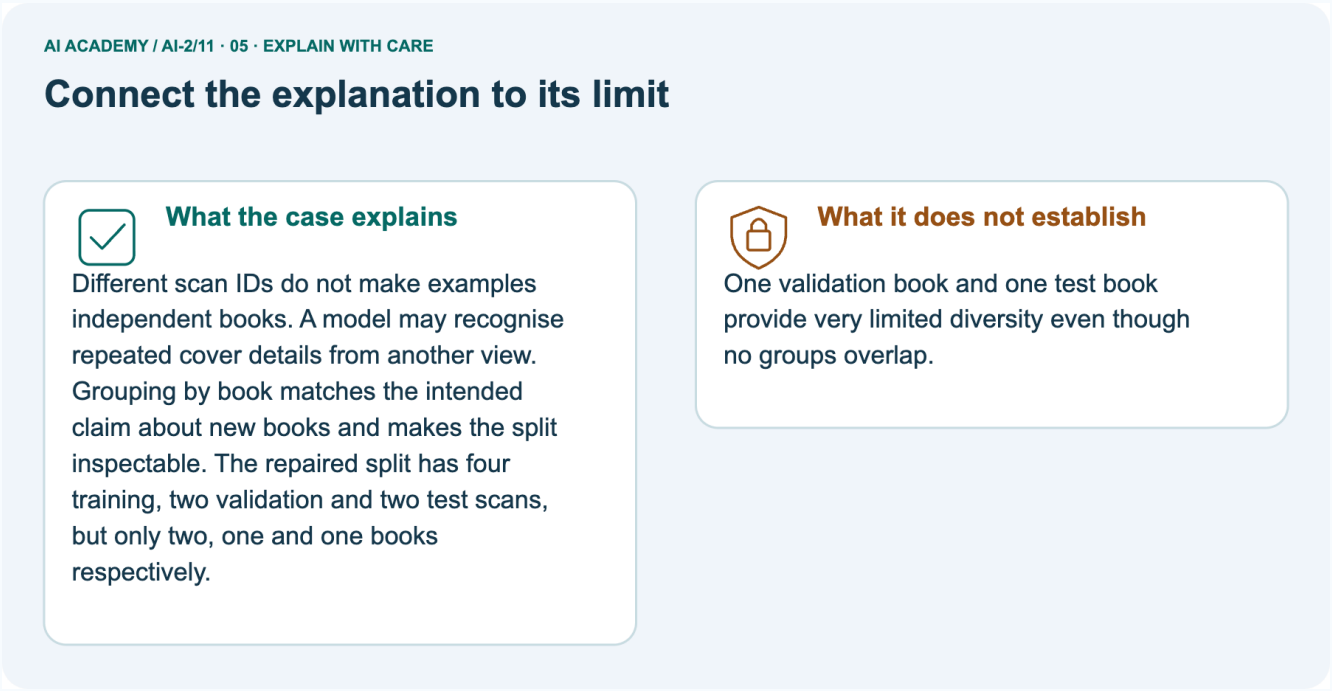
Original case: Keep every scan from one book in exactly one role before **fitting**. Changed condition: Give A2 a new filename without changing the book it depicts.

Expected result: Its **group** is still A; the original related examples still cross the split. Reason: Renaming a row does not remove the relationship between scans.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

My picture explanation, deciding evidence and one limit

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

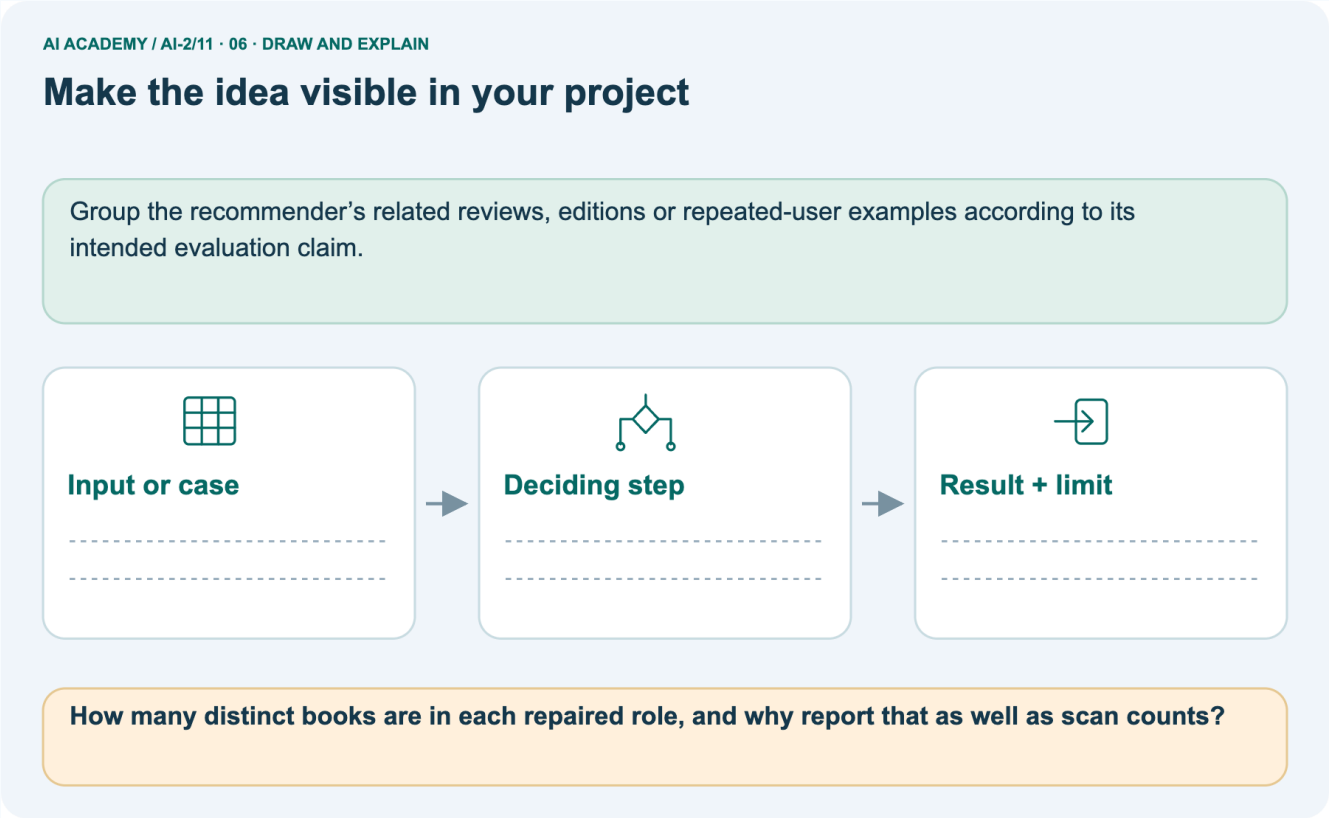
Different scan IDs do not make examples independent books. A model may recognise repeated cover details from another view. **Group**ing by book matches the intended claim about new books and makes the split inspectable.

The repaired split has four **training**, two validation and two test scans, but only two, one and one books respectively. Limit: One validation book and one test book provide very limited diversity even though no **group**s overlap.

Explain the case, then point to a limit on the claim.

My picture explanation, deciding evidence and one limit

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

Group the recommender's related reviews, editions or repeated-user examples according to its intended evaluation claim. Draw your **input**, deciding step, result and limit.

Supported question: How many distinct books are in each repaired role, and why report that as well as scan counts?

Draw your project trace and justify the deciding step using **evidence**.

My picture explanation, deciding evidence and one limit

▼ Earlier wording, alternatives and picture transcripts

01. Training and Testing: the case

02. Follow the deciding steps

03. Keep each trip in one role

04. Change one thing. Trace what follows.

05. Connect the explanation to its limit

06. Make the idea visible in your project

▼ Optional interactive practice and completed work

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.